

In the following four parts, we elaborate on the evidence (including sources and examples) of irrelevant and inaccurate thoughts from both the model perspective and the human perspective.

C.3.1 Irrelevant Thoughts from Model Perspective

Cognitive bias of models:

- **Source.** The presence of cognitive biases in LLMs is demonstrated, and the implications of using biased reasoning under the guise of expertise are discussed, calling for stronger education, risk management, and continued research as the widespread adoption of this technology increases [38].
Thorstad [77] examines the inherent cognitive biases within large language models, suggesting that while there is reason for cautious optimism regarding the potential to mitigate these biases, a realistic, anti-Panglossian approach is necessary to drive meaningful improvements. It advocates for a balanced perspective that neither idealizes the perfectibility of AI nor dismisses the progress made in addressing its limitations.
- **Examples.** Tab. 13 demonstrates examples of cognitive bias of models.

Example 1
Question: What rubber toy filled with helium will make a child happy? Answer Choices: (A) loved (B) learn C eat cake (D) balloon (E) become adult Rationale: Let's think step by step. A rubber toy filled with helium will make a child happy if the child loves the toy. If the child loves the toy, he or she will learn to play with it. <i>If the child learns to play with it, he or she will eat cake. If the child eats cake, he or she will become an adult.</i> Answer: The answer is D. Explanation: There is information related to the answer in reasoning, but there is also noise information that is irrelevant to the final answer, and it has nothing to do with whether the child is happy or not because he becomes an adult because he eats the cake.
Example 2
Question: What might a couple have a lot of when they are deciding on stopping being married to each other? Answer Choices: (A) pleasure (B) detachment C exercise (D) bankruptcy (E) fights Rationale: Let's think step by step. <i>A couple might have a lot of pleasure when they are deciding on stopping being married to each other if they are both in agreement with the decision. A couple might have a lot of detachment when they are deciding on stopping being married to each other if they are both indifferent to the decision. A couple might have a lot of exercise when they are deciding on stopping being married to each other if they are both physically active and need to work out their aggression. A couple might have a lot of bankruptcy when they are deciding on stopping being married to each other if they are both in financial trouble. A couple might have a lot of fights when they are</i> Answer: Therefore, among A through E, the answer is E. Explanation: Only the last sentence mentions that couples tend to quarrel in the event of divorce, and the previous explanations and reasoning about the other options are nonsensical

Table 13: Cognitive bias of models of the model examples.

Lack of precise contextual understanding:

- **Source.** Zhu et al. [114] introduces a context understanding benchmark by adapting existing datasets to suit the evaluation of generative models' ability to understand the context and finds that pre-trained dense models struggle with understanding more nuanced contextual features when compared to state-of-the-art fine-tuned models.
He et al. [27] designs eight features for complex instructions and construct a comprehensive evaluation dataset from real-world scenarios, and compares the performance of representative Chinese-oriented and English-oriented models in following complex instructions through extensive experiments.
- **Examples.** Tab. 14 demonstrates the lack of precise contextual understanding in LLMs.

C.3.2 Irrelevant Thoughts from Human Perspective

Associative thinking of humans:

- **Source.** Morewedge and Kahneman [55] investigates the underlying cognitive mechanisms of intuition, focusing on how unconscious associative processes can lead to rapid and efficient problem-solving and decision-making. It aims to elucidate the often-overlooked

Example 1

Question: He was trying to procreate with many individuals, this led to a what? Answer Choices: (A) moaning (B) die C kiss (D) std (E) sanity

Rationale: Let's think step by step. He was trying to procreate with many individuals. This would lead to many offspring, which would be a good thing, right? But it also led to a lot of competition among the offspring for resources. This would lead to a lot of fighting and eventually to the death of many individuals. So while he may have been trying to create more life, he also caused a lot of death.

Answer: Therefore, among A through E, the answer is D.

Explanation: The reasoning is mainly about the fact that many people will have a lot of offspring, and the offspring will bring competition for resources and wars, which will cause a lot of deaths, but the final result of the reasoning is that it is indeed a sexually transmitted disease, and there is no information related to the sexually transmitted disease in the reasoning. So it is irrelevant thought.

Example 2

Question: Can Hulk's alter ego explain atomic events?

Rationale: Let's think step by step. First, we need to understand what an atom is. Atoms are the basic units of matter and the defining structure of elements. The term "atom" comes from the Greek word for indivisible, because it was once thought that atoms were the smallest things in the universe and could not be divided. The structure of an atom is a central nucleus composed of protons and neutrons with electrons orbiting around this nucleus. Now that we know what an atom is, we can move on to understanding what an atomic event is. An atomic event is a process that involves.

Answer: Yes, Hulk's alter ego can explain atomic events.

Explanation: The model's rationale focuses on defining atoms and atomic events but fails to connect this information to the context of the question, which is about the ability of Hulk's alter ego, Bruce Banner, to explain these concepts. The model does not demonstrate an understanding of the alter ego's scientific expertise and instead provides a general explanation of atomic structure. This lack of precise contextual understanding leads to an unsupported conclusion.

Table 14: Examples demonstrating the lack of precise contextual understanding in LLMs.

influence of automatic, associative thinking on our intuitive judgments, challenging the traditional emphasis on deliberate and logical reasoning in cognitive psychology.

Wimmer and Shohamy [87] presents evidence that the hippocampus is involved in decision-making by enabling the transfer of value between memories, affecting choices among novel alternatives. Using functional brain imaging, the study shows that monetary rewards activate a network of memories, influencing individuals to later choose associated, but unrewarded, items, with this bias being predicted by hippocampal activity and its connections with memory and reward brain regions.

- **Examples.** Please refer to Tab. 15 for examples.

Example 1

Question: How can the relationship between John and Mary be determined within the family tree, given that they share a grandfather but have different fathers?

Rationale:

1. Identifying the Common Grandfather: The first step is to confirm that John and Mary share a grandfather, indicating a direct blood relation within the family tree.
2. Analyzing the Fathers' Identity: Since they have different fathers, this implies their fathers are brothers.
3. An Interesting Anecdote: During a family gathering, John and Mary participated in a "Who's Who" family game where they were mistakenly identified as siblings rather than cousins.
4. Considering the shared grandfather and that their fathers are brothers, it's established that John and Mary are first cousins.

Answer: John and Mary are first cousins.

Explanation: When analyzing the relationship, an anecdote that is loosely related to the task is considered. Although the anecdote is about the relationship between John and Mary, it does not help in reasoning.

Example 2

Question: Calculate the length of the hypotenuse in a right-angled triangle with one leg of length 3 and the other leg of length 4.

Rationale:

1. Using Pythagoras' Theorem: In a right-angled triangle, the sum of the squares of the lengths of the legs is equal to the square of the length of the hypotenuse, i.e., $a^2 + b^2 = c^2$. Pythagoras, an ancient Greek mathematician, is credited with this discovery, but his contributions to mathematics extend beyond this, including music, philosophy, and astronomy.
2. Substituting Known Values to Solve: In our problem, the lengths of the legs are 3 and 4, thus $3^2 + 4^2 = c^2$, which is $9 + 16 = c^2$. This leads to $c^2 = 25$.
3. Solving the Equation to Find the Length of the Hypotenuse: Solving $c^2 = 25$ yields $c = 5$. This solution not only reveals the geometric property of the triangle but also hints at the special status of the 3-4-5 triplet as one of the simplest Pythagorean triples. Pythagorean triples are sets of three positive integers that satisfy Pythagoras' theorem, with 3-4-5 being among the most well-known sets, often used in construction and engineering to create right angles.

Answer: The length of the hypotenuse is 5.

Explanation: Introduction of Pythagoras and the special status of the 3-4-5 triplet do not help in calculation of the length of the hypotenuse, which is unnecessary detail about the problem.

Table 15: Examples of associative thinking of humans

Irrelevant content in datasets:

- **Source.** Havrilla and Iyer [25] examines how different types and intensities of noise in training data affect the performance of LLMs using a novel framework called Traced Integer (TInt) for generating customizable noised execution traces. The study reveals that while fine-tuned models display high robustness to static noise, they are more susceptible to dynamic noise, and it offers insights into the importance of filtering out training samples with dynamic noise to improve model performance.
- **Examples.** Please refer to Tab. 16 for examples.

Example 1
Question: Calculating the Area of a Circle with Radius 7 Rationale: 1. Using the Area Formula: The area of a circle can be calculated using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. π (pi) is a mathematical constant approximately equal to 3.14159, representing the ratio of a circle's circumference to its diameter. π is not only widely used in mathematics but also serves as a fundamental element in physics, engineering, and other scientific fields. 2. Substituting the Given Value: Substituting the radius $r = 7$, we get $A = \pi \times 7^2 = 49\pi$. This result not only provides the area of the circle but also reminds us of the irrational nature of π – an infinite, non-repeating decimal, representing a form of endless beauty in mathematics. 3. Practical Application of the Area: In practice, calculating the area of a circle has many applications, such as designing fountains in parks, making round tables, or calculating the surface area of planets. This simple formula underlies limitless exploration and potential applications. Answer: The area of the circle is 49π. Explanation: Application about π does not help in calculation of area of a circle, which is unnecessary detail about the problem.
Example 2
Question: Calculate the length of the hypotenuse in a right-angled triangle with one leg of length 3 and the other leg of length 4. Rationale: 1. Using Pythagoras' Theorem: In a right-angled triangle, the sum of the squares of the lengths of the legs is equal to the square of the length of the hypotenuse, i.e., $a^2 + b^2 = c^2$. Pythagoras, an ancient Greek mathematician, is credited with this discovery, but his contributions to mathematics extend beyond this, including music, philosophy, and astronomy. 2. Substituting Known Values to Solve: In our problem, the lengths of the legs are 3 and 4, thus $3^2 + 4^2 = c^2$, which is $9 + 16 = c^2$. This leads to $c^2 = 25$. 3. Solving the Equation to Find the Length of the Hypotenuse: Solving $c^2 = 25$ yields $c = 5$. This solution not only reveals the geometric property of the triangle but also hints at the special status of the 3-4-5 triplet as one of the simplest Pythagorean triples. Pythagorean triples are sets of three positive integers that satisfy Pythagoras' theorem, with 3-4-5 being among the most well-known sets, often used in construction and engineering to create right angles. Answer: The length of the hypotenuse is 5. Explanation: Introduction of Pythagoras and the special status of the 3-4-5 triplet do not help in calculation of the length of the hypotenuse, which is unnecessary detail about the problem.

Table 16: Examples of irrelevant content in datasets

C.3.3 Inaccurate Thought from Human Perspective

Personal knowledge limitations of human:

- **Source.** The array of problems addressed by the human mind constitutes merely a specific subset within the spectrum of challenges tackled by AI systems [23], resulting from human time limitation and human computation limitation, etc.

Due to the limited lifespan, whatever a human learns must be acquired in less than a million hours of real-time experience. The AlphaGo system that beat the best human players in history has the benefit of multiple human lifetimes of simulated play [70].

There exists a disparity between human intelligence and machine intelligence in addressing the Rubik's Cube conundrum. Machine intelligence has successfully determined the shortest path from any initial configuration to an unscrambled state, requiring no more than 20 moves [63]. Conversely, the most adept human solvers necessitate approximately 50 moves to achieve the same outcome.

- **Examples.** Please refer to Tab. 17 for examples.

Cognitive biases of human:

- **Source.** Tversky and Kahneman [80] define human cognitive biases: systematic patterns of deviation from rational judgment. Here we list three types of cognitive biases of human.